

Boosting Ensembles for Improved Predictive Performance

Agenda

- Boosting
- Bagging Vs Boosting
- AdaBoost
- Gradient Boost
- XGBoost

Let's begin the discussion by answering a few questions

What is a key differentiation of boosting algorithms from bagging algorithms?

A

Boosting trains all models in parallel

B

Boosting decreases model variance

C

Boosting increases model interpretability

D

Boosting trains all models sequentially to improve upon the wrong predictions of the previous weak learner

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Bagging vs Boosting

Bagging

Creates different subsets of data using random sampling with replacement to independently train each weak learner

All weak learners are independent and can be trained in parallel

Aims to decrease variance, not bias

E.g, Random Forests

Boosting

Trains the weak learners sequentially on the entire dataset, reweighting misclassified data points to focus on reducing errors iteratively

All weak learners are dependent on each other and have to be trained sequentially

Aims to decrease bias, not variance

E.g, AdaBoost, Gradient Boost

Boosting Quiz

Statement 1: Boosting trains weak learners sequentially, increasing the weights of misclassified data points so that subsequent models focus more on difficult samples.

Statement 2: Boosting trains all weak learners independently in parallel, and each data point is given equal importance throughout the training process.

Statement 3: Boosting always reduces training time because it combines all weak learners into a single model at once.

Which of the following is correct?

A

Statement 1

B

Statement 2

C

Statement 1,2

D

Statement 3

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Boosting

Boosting is an ensemble learning technique that combines multiple weak learners to form a strong model.

It trains models sequentially, where each new learner focuses more on the mistakes made by the previous ones.

Misclassified data points are assigned higher weights, while correctly classified points receive lower weights. This process continues until a stopping criterion such as maximum iterations or desired performance is reached.

The final prediction is made using weighted averaging or weighted voting of all learners

In AdaBoost, how does the algorithm adjust the weights of samples that are misclassified by a weak learner?

A

Increased

B

Decreased

C

Randomly assigned

D

Remains unchanged

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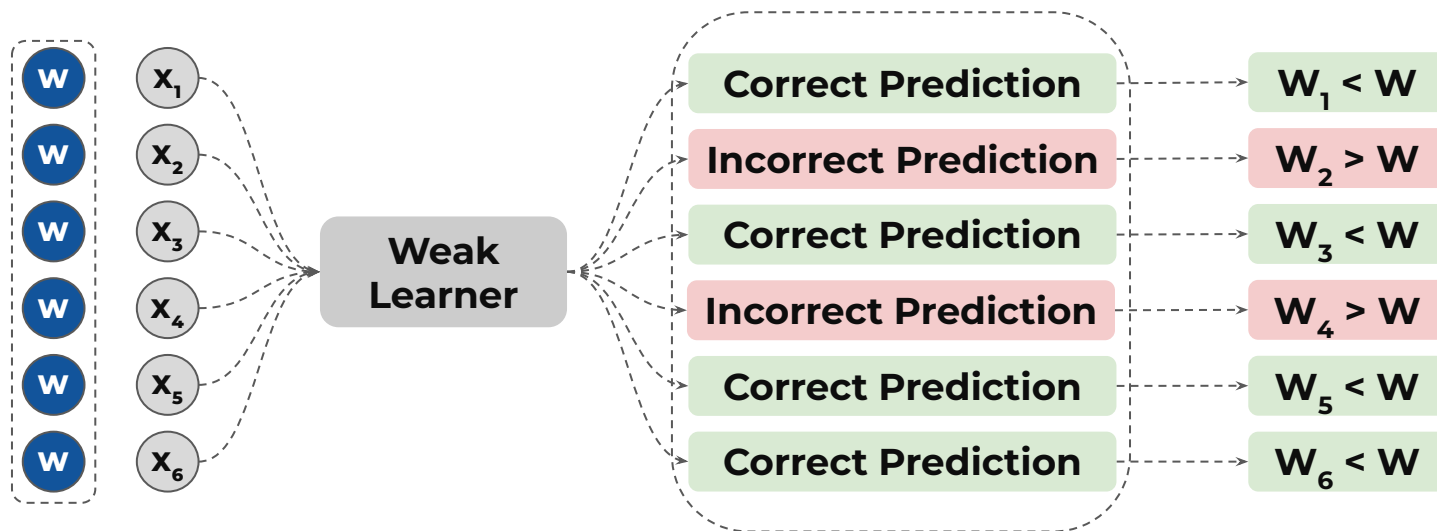
Randomly assigned

D

Remains unchanged

AdaBoost - Weight Updates

Weights are **increased** for the **misclassified instances** and **decreased** for the **correctly classified instances**



What role does the learning rate play in Gradient Boosting Models?

A

Adjusts the depth of trees

B

Controls the contribution of each tree to the final ensemble

C

Defines the number of weak learners to be trained

D

Determines the fraction of data used

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Learning Rate in Gradient Boost

Learning rate (η) is a **small positive number** that **controls the contribution of each weak learner** to the ensemble

It controls the **magnitude of the steps** taken by the ensemble towards **an optimal solution**

0.0

η

∞

$$\hat{Y}_i = \hat{Y}_{i-1} + \eta \cdot h_i$$

η is **generally** set to smaller values (**0.01 to 0.2**) to take small steps towards the optimal solution and avoid overfitting

The **learning rate** is **multiplied** with the **predicted residual (h_i)** from each weak learner **added to the final ensemble**, thereby **dictating their contribution** by **adjusting the magnitude** of the predicted residual

What is the primary difference between the ways Gradient Boost and AdaBoost improve their performance with each iteration?

A

Gradient Boosting minimizes the residual errors, while AdaBoost updates sample weights

B

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C

Both methods update sample weights

D

Both methods minimize residual errors

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AdaBoost V/S Gradient Boost

AdaBoost

Begins with equal importance for all data points

After training, the algorithm increases importance of misclassified instances

Subsequent learners focus more on these challenging instances

The final model combines weak learners by assigning higher importance to learners that perform well on challenging instances

Gradient Boost

Begins with an initial learner that makes simple predictions, like log(odds) value or a simple decision tree

After training, the algorithm focuses on minimizing residuals (difference between observed and predicted values)

Subsequent learners focus on learning patterns in the residuals from the previous learners

The final model combines weak learners by adding them sequentially, with each new learner reducing the errors made by the previous ones

Boosting Quiz

In XGBoost, the parameter _____ controls the step size shrinkage used to prevent overfitting, _____ specifies the minimum loss reduction required to make a further partition on a leaf node, and _____ is used to handle class imbalance by controlling the balance between positive and negative classes.

A

`learning_rate, gamma, scale_pos_weight`

B

`gamma, learning_rate, scale_pos_weight`

C

`gamma, learning_rate, scale_pos_weight`

D

`scale_pos_weight, gamma, learning_rate`

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`scale_pos_weight, gamma, learning_rate`

Parameters of XGBoost

Parameters	Description
learning_rate	Gradient boosted decision trees are quick to learn and overfit training data. One effective way to slow down learning in the model is to use a learning rate, also called shrinkage
gamma	A node is split only when the resulting split gives a positive reduction in the loss function. Gamma specifies the minimum loss reduction required to make a split. Higher the gamma value, the lesser the chances of overfitting
scale_pos_weight	Control the balance of positive and negative weights. It can have any positive value as input. It helps in imbalanced classification problems
colsample_bytree	Proportion of features randomly sampled to train each individual tree in XGBoost
colsample_bylevel	Fraction of features randomly sampled at each tree level when building an XGBoost model



Power Ahead !

